

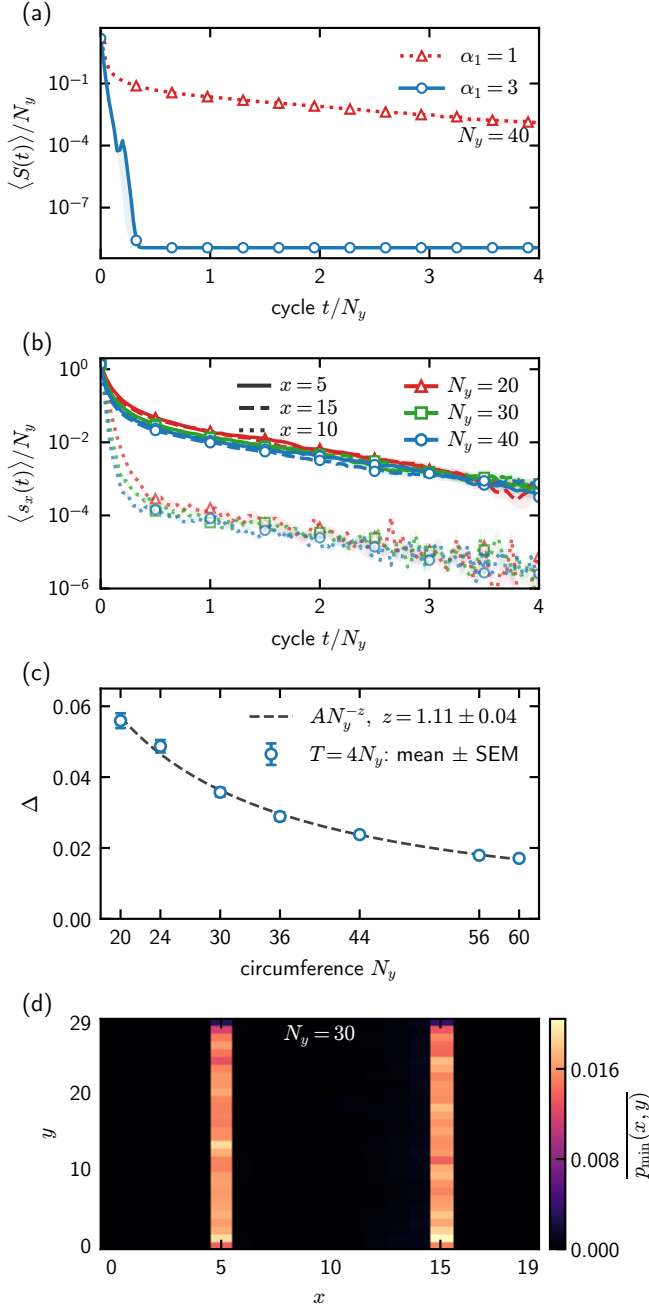


FIG. 1. Hard-wall purification from a maximally mixed active slab: $N_x = 20$, $\alpha_2 = 30$, $n_{\text{shell}} = 1$, perfect correction, 100 independent Born trajectories per configuration, and $T = 4N_y$. (a) Mean entropy at $N_y = 40$ for $\alpha_1 = 1, 3$. The near-zero control tail is the numerical entropy floor. (b) Column contours at $\alpha_1 = 1$, summed over y and orbitals, for $N_y = 20, 30, 40$. Bands denote one sample SEM. (c) Mean endpoint gap $\Delta = \min_j |\lambda_j|$, with $\lambda_j = -\text{atanh}(a_j)/T$, centered covariance eigenvalues a_j , and SEM bars. The all-size power-law fit gives $\Delta \propto N_y^{-1.11 \pm 0.04}$. (d) At $N_y = 30$, $\alpha_1 = 1$, the orbital-summed probability density of the single smallest- $|\lambda_j|$ eigenmode is selected separately in each trajectory, then averaged over 100 trajectories. All selected minima are non-degenerate. The mean density sums to one; colors are linear.